

MSDM5004

Numerical Methods and Modeling in Science
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Lecture 9

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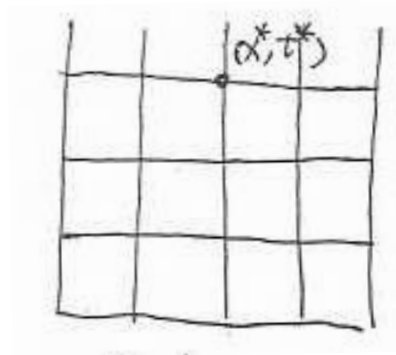
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1.3 Convergence

A numerical scheme is convergent if

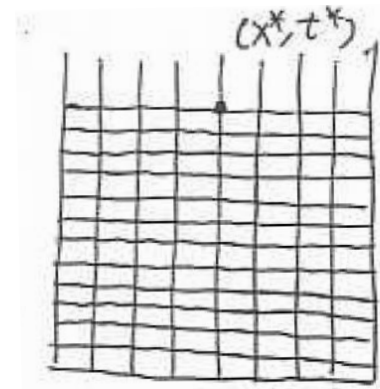
for any fixed point $(x^*, t^*) \in (0, 1) \times (0, t_F)$

$$U_j^n \rightarrow u(x^*, t^*) \quad \text{as } \Delta x \rightarrow 0, \Delta t \rightarrow 0 \text{ and } x_j \rightarrow x^*, t_n \rightarrow t^*.$$



$$\Delta x_1, \Delta t_1 \quad \begin{aligned} x^* &= j_1 \Delta x_1 \\ t^* &= n_1 \Delta t_1 \end{aligned}$$

$$U_{j_1}^{n_1} \approx u(x^*, t^*)$$



Smaller $\Delta x, \Delta t$

$$\begin{aligned} \Delta x_2 &= \frac{1}{2} \Delta x_1 & x^* &= j_2 \Delta x_2 \\ \Delta t_2 &= \frac{1}{4} \Delta t_1 & t^* &= n_2 \Delta t_2 \end{aligned}$$

$$U_{j_2}^{n_2} \approx u(x^*, t^*) \quad j_2 = 2j_1$$

$$n_2 = 4n_1$$

Convergence of the explicit method

Theorem

The explicit scheme
$$\frac{U_j^{n+1} - U_j^n}{\Delta t} = \frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{(\Delta x)^2}$$

is convergent if

$$\mu = \frac{\Delta t}{(\Delta x)^2} \leq \frac{1}{2}.$$

Proof

Define the error $e_j^n = U_j^n - u(x_j, t_n)$.

The numerical scheme is

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} = \frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{(\Delta x)^2}$$

The exact solution satisfies

$$\frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} = \frac{u(x_{j+1}, t_n) - 2u(x_j, t_n) + u(x_{j-1}, t_n)}{(\Delta x)^2} = T(x_j, t_n)$$

Subtract the second equation from the first one,

$$\frac{e_j^{n+1} - e_j^n}{\Delta t} - \frac{e_{j+1}^n - 2e_j^n + e_{j-1}^n}{(\Delta x)^2} = -T_j^n \quad (T_j^n = T(x_j, t_n))$$

Multiplying both sides by Δt and using $\mu = \frac{\Delta t}{(\Delta x)^2}$,

$$\begin{aligned}e_j^{n+1} &= e_j^n + \mu (e_{j+1}^n - 2e_j^n + e_{j-1}^n) - \Delta t T_j^n \\ &= \mu e_{j+1}^n + (1 - 2\mu)e_j^n + \mu e_{j-1}^n - \Delta t T_j^n.\end{aligned}$$

Thus, if $\mu \leq \frac{1}{2}$,

$$\begin{aligned}|e_j^{n+1}| &\leq \mu |e_{j+1}^n| + |1 - 2\mu| |e_j^n| + \mu |e_{j-1}^n| + \Delta t |T_j^n| \\ &= \mu |e_{j+1}^n| + (1 - 2\mu) |e_j^n| + \mu |e_{j-1}^n| + \Delta t |T_j^n|\end{aligned}$$

Let
$$E^n = \max_{0 \leq j \leq J} |e_j^n|,$$

$$\bar{T} = \max_{0 \leq j \leq J, n\Delta t \leq t_F} |T_j^n| = \frac{1}{2}M_{tt}\Delta t + \frac{1}{12}M_{xxxx}(\Delta x)^2$$

Then

$$\begin{aligned} |e_j^{n+1}| &\leq \mu|E^n| + (1 - 2\mu)|E^n| + \mu|E^n| + \Delta t\bar{T} \\ &= E^n + \Delta t\bar{T}. \end{aligned}$$

Since this inequality holds for all j , we have

$$E^{n+1} \leq E^n + \Delta t\bar{T}.$$

Therefore,

$$\begin{aligned} E^n &\leq E^{n-1} + \Delta t \bar{T} \\ &\leq (E^{n-2} + \Delta t \bar{T}) + \Delta t \bar{T} \\ &= E^{n-2} + 2\Delta t \bar{T} \\ &\dots \\ &\leq E^0 + n\Delta t \bar{T} \\ &\leq t_F \bar{T} \quad (E^0 = 0) \\ &= t_F \left[\frac{1}{2} M_{tt} \Delta t + \frac{1}{12} M_{xxxx} (\Delta x)^2 \right] \\ &\rightarrow 0, \quad \text{as } \Delta x, \Delta t \rightarrow 0. \end{aligned}$$

This means that the numerical scheme is convergent.

If $\mu = \frac{\Delta t}{(\Delta x)^2}$ is constant,

$$|U_j^n - u(x_j, t_n)| = |e_j^n| \leq E^n \leq t_F \left[\frac{1}{2} M_{tt} \mu + \frac{1}{12} M_{xxxx} \right] (\Delta x)^2$$

Refinement path

A sequence $\{(\Delta x_i, \Delta t_i), i = 0, 1, 2, \dots, \Delta x_i, \Delta t_i \rightarrow 0\}$

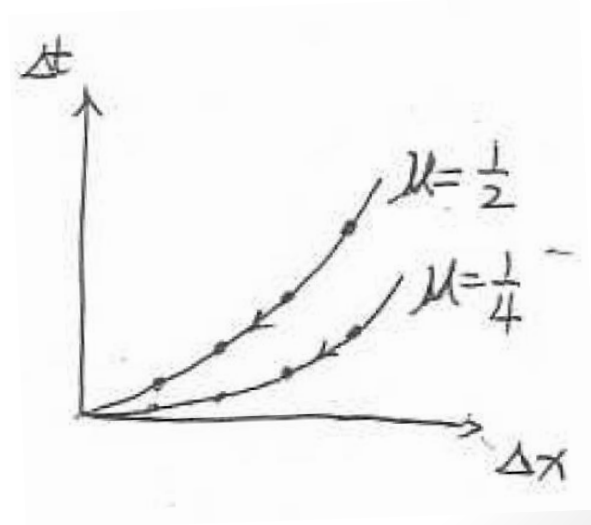
$$\text{If } \mu_i = \frac{\Delta t_i}{(\Delta x_i)^2} \leq \frac{1}{2},$$

then $\{U_{j_i}^{n_i}\}, i = 0, 1, 2, \dots$, converges to $u(x^*, t^*)$,

where $x^* = j_i \Delta x_i, t^* = n_i \Delta t_i$.

Commonly used:

$$\mu = \frac{\Delta t}{(\Delta x)^2} = \text{constant} \leq \frac{1}{2}.$$



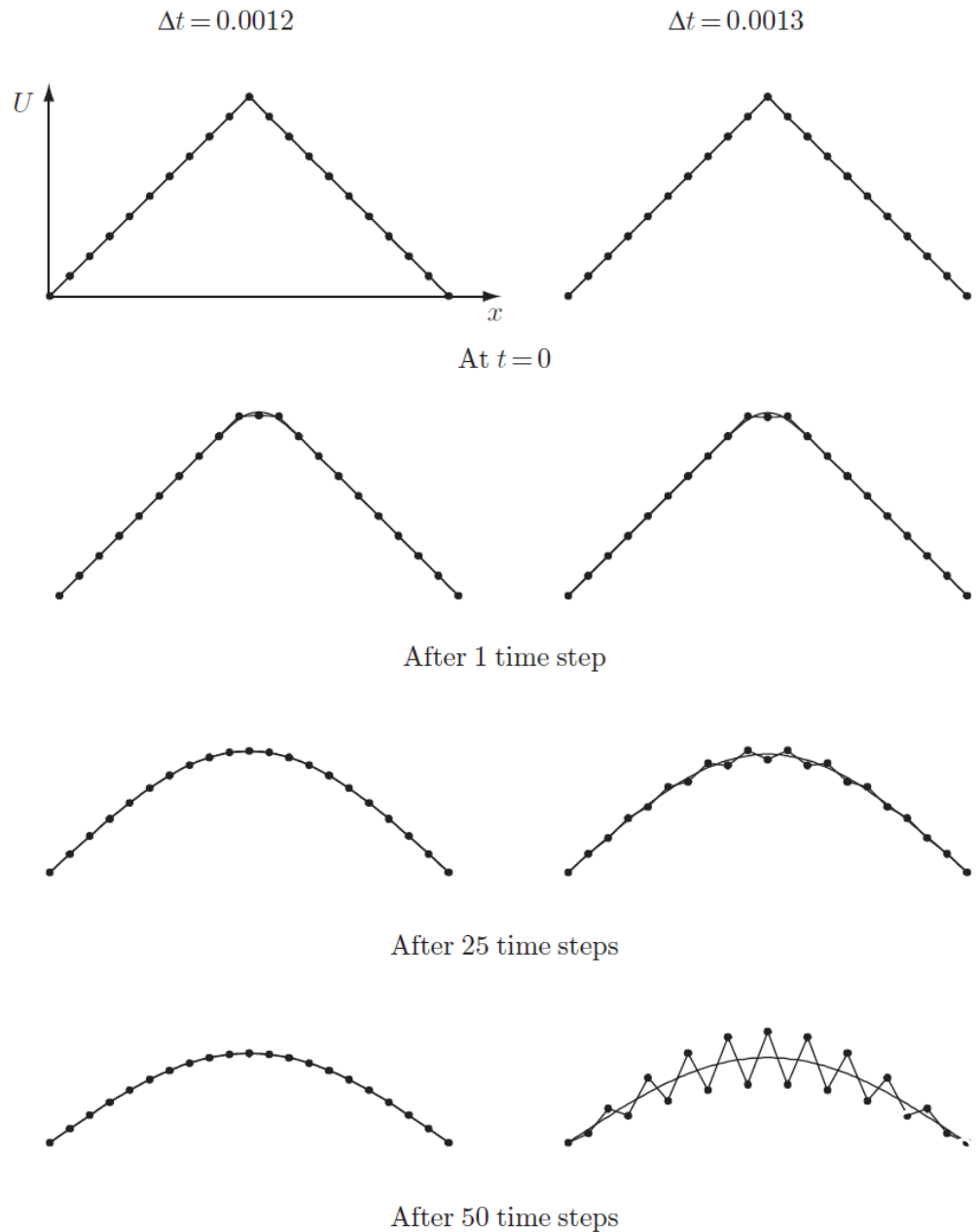
An example

$$u^0(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq \frac{1}{2}, \\ 2 - 2x & \text{if } \frac{1}{2} \leq x \leq 1. \end{cases}$$

$$J = 20, \Delta x = 0.05.$$

$$\text{When } \mu = \frac{\Delta t}{(\Delta x)^2} = \frac{1}{2}$$

$$\Delta t = 0.00125$$



Remark

The exact solution of the problem

$$\frac{\partial u}{\partial t}(x, t) = \frac{\partial^2 u}{\partial x^2}(x, t), \quad 0 < x < 1, \quad t > 0.$$

$$u(0, t) = u(1, t) = 0, \quad t > 0$$

$$u(x, 0) = u_0(x), \quad 0 \leq x \leq 1.$$

is (can be found by method of separation of variables)

$$u(x, t) = \sum_{m=1}^{\infty} a_m e^{-(m\pi)^2 t} \sin m\pi x$$

$$a_m = 2 \int_0^1 u^0(x) \sin m\pi x \, dx.$$

1.4. Stability

A numerical scheme (for a linear PDE) is stable if

$$\|\mathbf{U}^n\| \leq K\|\mathbf{U}^0\| \quad (\text{any norm})$$

where K is a constant.

Here $\mathbf{U}^n$ denotes the vector formed by $\{U_j^n\}$

Fourier analysis

For the numerical scheme

$$U_j^{n+1} = U_j^n + \mu(U_{j+1}^n - 2U_j^n + U_{j-1}^n) \quad \mu = \frac{\Delta t}{(\Delta x)^2}$$

Assume $U_j^n = [\lambda(k)]^n e^{ik(j\Delta x)}$

$\lambda(k)$ is a number depending on k

Substitute it into the scheme

$$[\lambda(k)]^{n+1} e^{ik(j\Delta x)} = [\lambda(k)]^n e^{ik(j\Delta x)} + \mu \left([\lambda(k)]^n e^{ik((j+1)\Delta x)} - 2[\lambda(k)]^n e^{ik(j\Delta x)} + [\lambda(k)]^n e^{ik((j-1)\Delta x)} \right)$$

Dividing both sides by $[\lambda(k)]^n e^{ik(j\Delta x)}$

$$\begin{aligned}\lambda(k) &= 1 + \mu (e^{ik\Delta x} - 2 + e^{-ik\Delta x}) \\ &= 1 - 2\mu(1 - \cos k\Delta x) \\ &= 1 - 4\mu \sin^2 \frac{1}{2}k\Delta x;\end{aligned}$$

$\lambda(k)$ is called the *amplification factor* for the mode.

A numerical scheme (for a linear PDE) is stable if

$$|[\lambda(k)]^n| \leq K, \quad \text{for } n\Delta t \leq t_F, \quad \forall k.$$

where K is a constant.

Since

$$0 \leq \sin^2 \frac{k\Delta x}{2} \leq 1$$

$$0 \leq 4\mu \sin^2 \frac{k\Delta x}{2} \leq 4\mu$$

$$1 - 4\mu \leq 1 - 4\mu \sin^2 \frac{k\Delta x}{2} \leq 1$$

$$\left| 1 - 4\mu \sin^2 \frac{k\Delta x}{2} \right| \leq \max\{1, |1 - 4\mu|\}$$

Note that the maximum may be achieved when $k = J\pi/L$

$$\sin \frac{k\Delta x}{2} = \sin \frac{\pi}{2} = 1$$

$$\left| 1 - 4\mu \sin^2 \frac{k\Delta x}{2} \right| = |1 - 4\mu|$$

$$\left| 1 - 4\mu \sin^2 \frac{k\Delta x}{2} \right|^n \text{ is bounded}$$

$$\text{if and only if } \left| 1 - 4\mu \sin^2 \frac{k\Delta x}{2} \right| \leq 1$$

$$\text{if and only if } |1 - 4\mu| \leq 1$$

$$\text{if and only if } \mu \leq \frac{1}{2}.$$

Therefore, the explicit scheme is stable if and only if

$$\mu \leq \frac{1}{2}.$$

For a general initial condition $u_0(x)$, we have

$$u_0(x) = \sum_k C_k e^{ikx}$$

Thus

$$u_j^0 = \sum_k C_k e^{ik(j\Delta x)}$$

Solution of a linear scheme has the form

$$u_j^n = \sum_k C_k [\lambda(k)]^n e^{ik(j\Delta x)}.$$

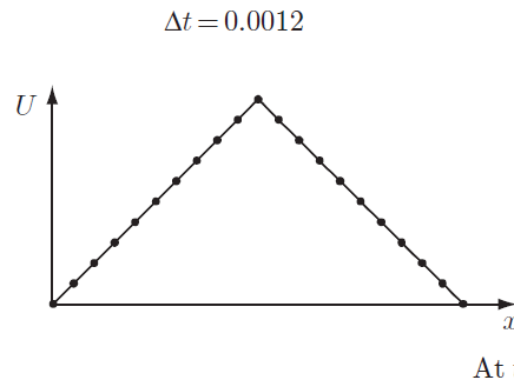
An example

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$$J = 20, \quad \Delta x = 0.05.$$

$$\text{When } \mu = \frac{\Delta t}{(\Delta x)^2} = \frac{1}{2}$$

$$\Delta t = 0.00125$$



After 1 time step

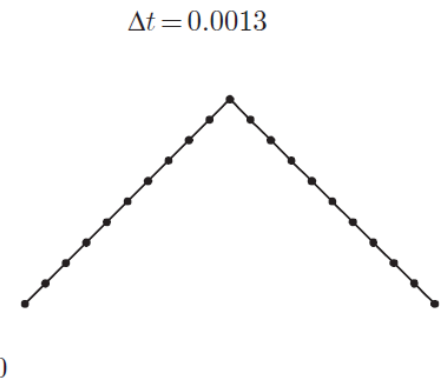


After 25 time steps



After 50 time steps

Stability condition
is satisfied



Stability condition
is not satisfied

$$U_j^n = [\lambda(k)]^n e^{ik(j\Delta x)}$$

e^{ikx} period L

k is the wavenumber, $k = \frac{2\pi}{L}m$ with integer m $e^{ikx} = e^{i2\pi m \frac{x}{L}}$

$\frac{2\pi}{k} = \frac{L}{m}$ is a period

For the explicit scheme $\lambda(k) = 1 - 4\mu \sin^2 \frac{1}{2}k\Delta x$

Stability condition $\mu \leq \frac{1}{2}$

If $\mu > 1$, maximum of λ is achieved when $k\Delta x = \pi$

The period in this case is $\frac{2\pi}{k} = \frac{2\pi}{\pi/\Delta x} = 2\Delta x$